Examiner only

7. (a) State **three** assumptions of the kinetic theory of ideal gases. [3]

(b) A container of fixed volume contains oxygen gas. Five molecules in the gas have speeds 480, 521, 436, 445 and 503 $m\,s^{-1}$.

(i) Determine the rms speed of these five molecules. [2]

(ii) Determine the temperature of a sample of oxygen gas whose molecules have this rms speed. (Relative molecular mass of oxygen = 32.) [3]

Examiner only

(c) Gareth states that when the temperature of the gas is doubled both the rms speed of the molecules and the pressure are doubled. Determine whether or not Gareth is correct. [4]

....

....

....

....

....

....

12